Open Source Tools that Assist Industry 5.0

Unlike Industry 4.0, Industry 5.0 integrates human expertise with AI. There are quite a few open source libraries that can be used for this new global system.

Industry 5.0 focuses on human-centric machine collaboration with artificial intelligence (AI) to enable real-time upgrades in a system. It does not replace human expertise but includes it so that better and effective AI robotic environments can be developed and deployed as per local needs.

Industry 4.0 involves complete automation and lacks human interaction. As a result, smart automation and robotic operations sometimes do not fit well in a particular environment. Industry 5.0 fills this gap because human experts and specialists take care of each phase in the robotic AI environment.

Table 1: Industry 5.0 vs Industry 4.0 in the context of robotic surgery

Aspect and integration feature	Industry 5.0	Industry 4.0
Role	Human guidance	Automation
Control	Surgeon driven	Machine driven
AI	Decision support	Task execution
Ethics	Central	Secondary
Outcome	Patient well-being	Efficiency

Key components of Industry 5.0 are:

- Adaptive networks
- Agentic AI
- Circular economy
- Cognitive automation
- Cyber-physical systems
- Data sovereignty
- Digital twins
- Ethical AI
- Green technologies
- Human-AI collaboration
- Human-centricity
- Industrial IoT
- Mass customisation
- Personalised production
- Resilient systems